Precision of β^+ distal-edge verification with a closed-ring CRYSP scanner

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Abstract

We quantify the precision with which an in-room PET scanner determines the distal edge of a proton field from the β^+ activity it induces. Using ten independent simulated 1 Gy acquisitions of an SOBP field in a head phantom, imaged by a 1 m-long closed BGO ring and reconstructed with the full list-mode chain, we establish the estimator — the midpoint R_{50} of the distal activity fall-off, fitted on the whole-plane depth profile — and measure its performance. The activity–dose offset is a calibration constant, determined by the reference simulation with a systematic budget of 0.45 mm (acceptance −0.25 mm, reconstruction −0.23 mm, uncorrected scatters 0.02 mm), each term resolved to 0.01 mm to 0.02 mm. A single 1 Gy acquisition measures the edge with $\sigma_R = 0.057(13)$ mm; the precision follows $1/\sqrt{\text{dose}}$ and the calibration is dose-invariant, so a 0.1 Gy acquisition still reaches $\sigma_R = 0.18$ mm. An exploratory shot at a small fraction of the therapeutic dose can therefore verify the beam range at the two-tenths-of-a-millimetre level before the clinical dose is delivered.

1 Setup

The study case is an SOBP proton field depositing a uniform 1 Gy in a head phantom: an ellipsoid of uniform brain tissue with semi-axes $(72, 87, 102)$ mm (Fig. 1, left, shows the beam; the beam axis is z). The nuclear interactions of the protons produce β^+ emitters along the beam path — mainly ^{15}O ($T_{1/2} = 122$ s) and ^{11}C (20.4 min), with ^{13}N, ^{10}C and ^{14}O contributions — whose annihilation photons the scanner images. Figure 1 (right) shows the two nominal curves of the problem: the depth–dose $D(z)$ and the depth profile of the β^+ activity $A(z)$. Because isotope production requires protons above the reaction thresholds (≈ 17 and 18 MeV for the main ^{15}O and ^{11}C channels), the activity ends about 11 mm upstream of the dose edge. That offset is stable beam physics: it is what a simulation calibrates and a measurement rides on.

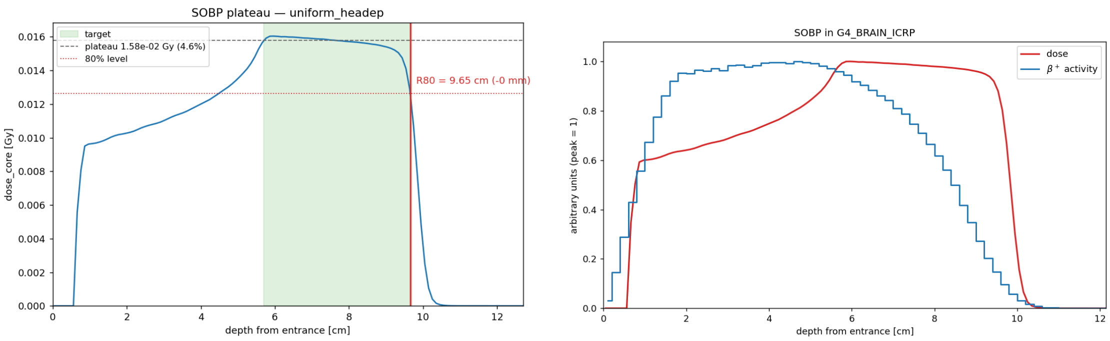

Figure 1: Left: the SOBP depth–dose and the definition of the clinical range point R_{80}. Right: the nominal β^+ activity profile together with the dose profile; the activity edge sits ≈ 11 mm upstream of the dose edge, set by the production thresholds.

The scanner is the reference configuration of the CRYSP programme [1]: a closed cylindrical ring of inner radius 387 mm and axial length 1024 mm, made of 960 monolithic BGO crystals ($51 \times 51 \times 37\,\text{mm}^3$, 3.3 radiation lengths), with spatial resolution $\sigma_{xyz} = 1.7$ mm, energy resolution 10% FWHM at 511 keV, a 450 keV acceptance threshold and a 3 ns coincidence window.

The acquisition covers 0 s to 1200 s after the irradiation. Ten statistically independent 1 Gy acquisitions were simulated; each records 17.4×10^6 coincidences, of which 81% are true, 18.7% scattered, and randoms are negligible.

Images are reconstructed with list-mode MLEM [2] on a $64 \times 64 \times 96$ grid of 1.5 mm voxels covering the activity corridor, with 50 iterations, per-event attenuation factors computed analytically on the phantom ellipsoid, and a sensitivity image estimated from 10^9 sampled lines of response. Figure 2 shows the orthogonal projections of one reconstructed acquisition: the activity corridor enters through the back of the head and terminates at the distal edge that the estimator measures.

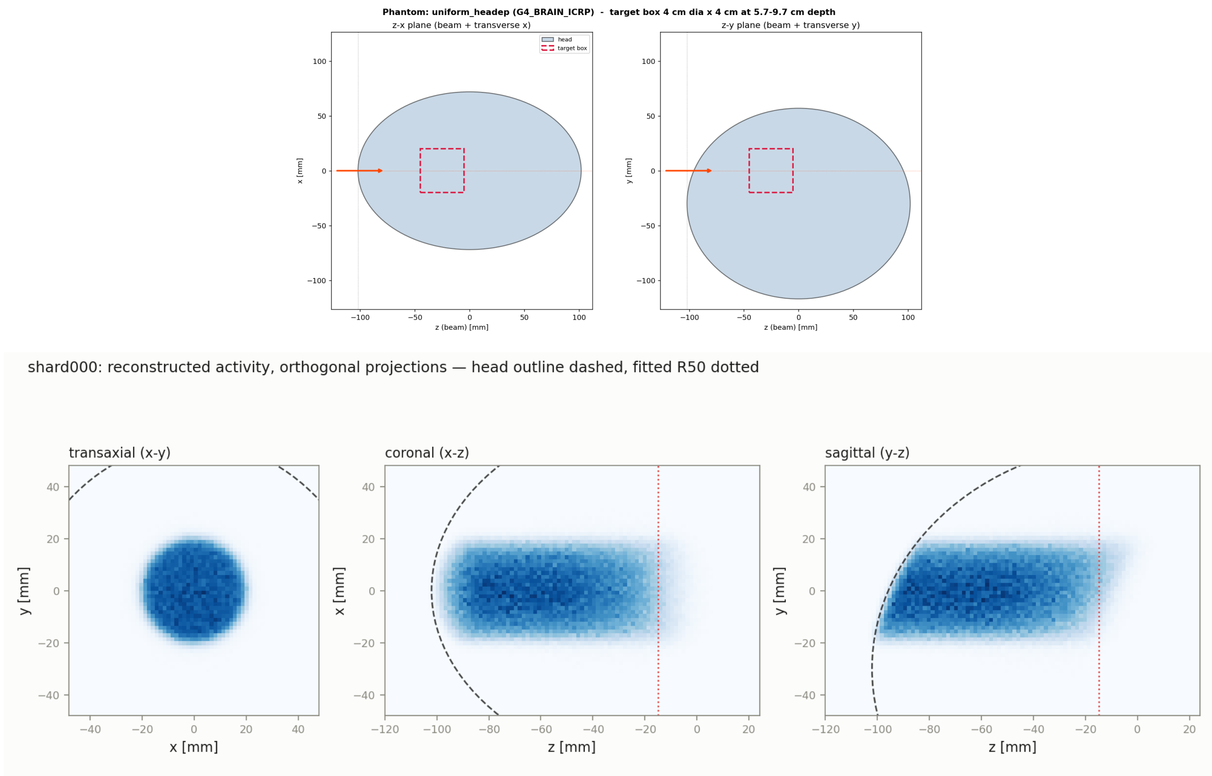

Figure 2: Top: the head phantom, an ellipsoid of uniform brain tissue. Bottom: orthogonal projections of the reconstructed activity of one 1 Gy acquisition (transaxial, coronal, sagittal). The dashed curve is the phantom outline; the dotted line marks the fitted R_{50}.

2 The estimator

The depth profile is the activity integrated over the full transverse plane at each depth,

$$P(z) \;=\; \iint A(x, y, z)\, \mathrm{d}x\, \mathrm{d}y, \tag{1}$$

which makes the profile insensitive to the transverse spreading of the beam with depth. The distal fall-off is fitted with a complementary error function plus a baseline,

$$P(z) \;=\; b + \frac{a}{2}\,\mathrm{erfc}\!\left(\frac{z - z_0}{\sqrt{2}\, w}\right), \tag{2}$$

inside a fixed window bracketing the edge ($-36.5 \leq z \leq -1.5$ mm for the nominal beam), with Poisson weights. The four parameters are the baseline b, the plateau amplitude a, the midpoint z_0 and the edge width w.

Range points. For any curve with a plateau and a distal fall-off, R_X denotes the depth at which the curve has dropped to $X\%$ of its plateau. Two of them play distinct roles:

- $R_{50} \equiv z_0$, the *midpoint* of the fall-off, is the range observable. It is where the profile is steepest, so it carries the smallest statistical error of any level, and Eq. (2) yields it directly as a fit parameter.

- R_{80} of the *dose* is the clinical prescription point (Fig. 1, left). For a given beam it differs from the dose R_{50} by a fixed geometric constant of the SOBP shape ($R_{80} - R_{50}^{\text{dose}} = -2.0\,\text{mm}$ for the nominal beam), so a measurement expressed in R_{50} converts to R_{80} exactly.

The ΔR estimator. The measured quantity is

$$\Delta R \;=\; R_{50}^{\text{activity}} - R_{50}^{\text{dose}}, \tag{3}$$

with the same fit construction, Eq. (2), applied to both curves. ΔR is calibrated once on the reference simulation of the treatment plan; a range deviation then appears as the difference between the measured R_{50}^{activity} and its calibrated expectation, and every constant of the convention cancels. For the nominal beam the truth-level value is $\Delta R = -10.74\,\text{mm}$ ($R_{50}^{\text{activity}} = -14.32(1)\,\text{mm}$, $R_{50}^{\text{dose}} = -3.58(3)\,\text{mm}$).

The estimator is insensitive to its own conventions: replacing the erfc by a logistic edge — the parametrisation of Ref. [3], whose positron-activity-range PAR is precisely R_{50} — moves R_{50} by a constant 0.10 mm at every rung of the analysis, and moves the calibrated differences by less than 0.02 mm. The fitted edge shape itself is a superposition: each isotope's activity terminates where the proton energy crosses its production threshold, so the fall-off is a sum of edges at slightly different depths — dominated by ^{15}O and ^{11}C, with a deeper-reaching ^{13}N foot from its low ($\approx 5.5\,\text{MeV}$) threshold. Equation (2) is thus a convention whose parameters are validated by their stability, and an isotope-resolved composite fit is a natural refinement (Sec. 6).

Figure 3 shows the fit on one reconstructed acquisition.

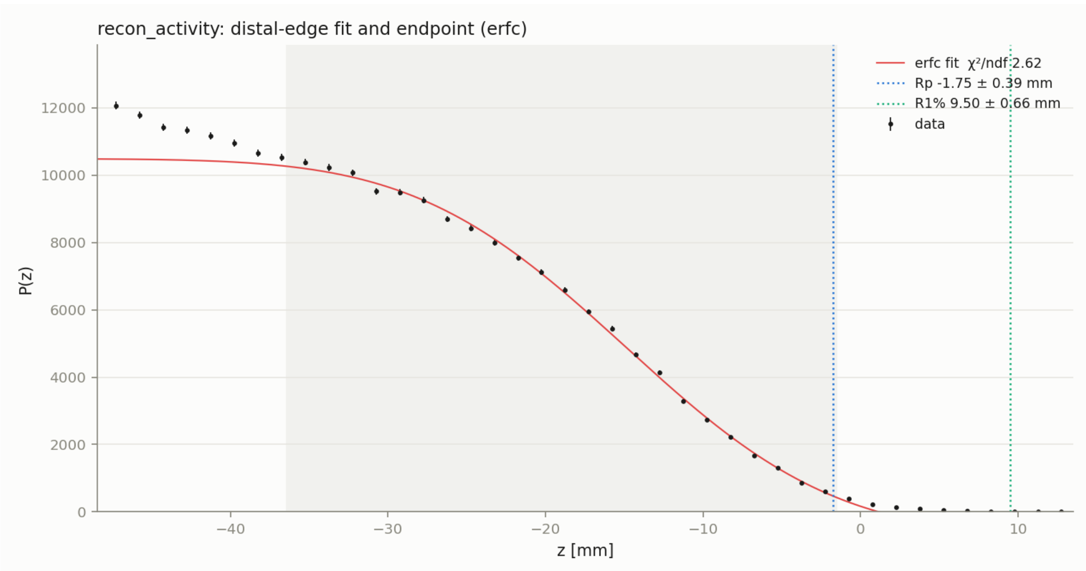

Figure 3: Whole-plane depth profile of one reconstructed 1 Gy acquisition (points, Poisson errors) and the erfc fit (Eq. 2, red line) in the fixed window (shaded). The markers show two derived points: the tangent endpoint R_p and the 1% crossing.

3 Calibration budget and precision at 1 Gy

Each stage of the measurement chain shifts ΔR by a definite, resolvable amount. Table 1 and Fig. 4 give the ladder, measured over the ten independent acquisitions: from the true activity to the origins of the detected true coincidences (the acceptance stage: the attenuation and solid-angle weighting of the detected sample), to the MLEM reconstruction of the trues, to the reconstruction of *all* events with no scatter or randoms correction.

Table 1: The calibration budget: ΔR at each rung of the measurement chain (erfc fit), over the ten 1 Gy acquisitions. The bracketed number is the acquisition-to-acquisition spread; the shift column is referred to the truth-level value −10.743 mm.

rung	ΔR [mm]	sem [mm]	[std, mm]	shift [mm]
truth activity	−10.743	—	—	—
detected trues (origins)	−10.988	0.010	0.031	−0.245
reconstruction (trues)	−11.216	0.018	0.057	−0.473
reconstruction (all events, uncorrected)	−11.194	0.022	0.071	−0.451

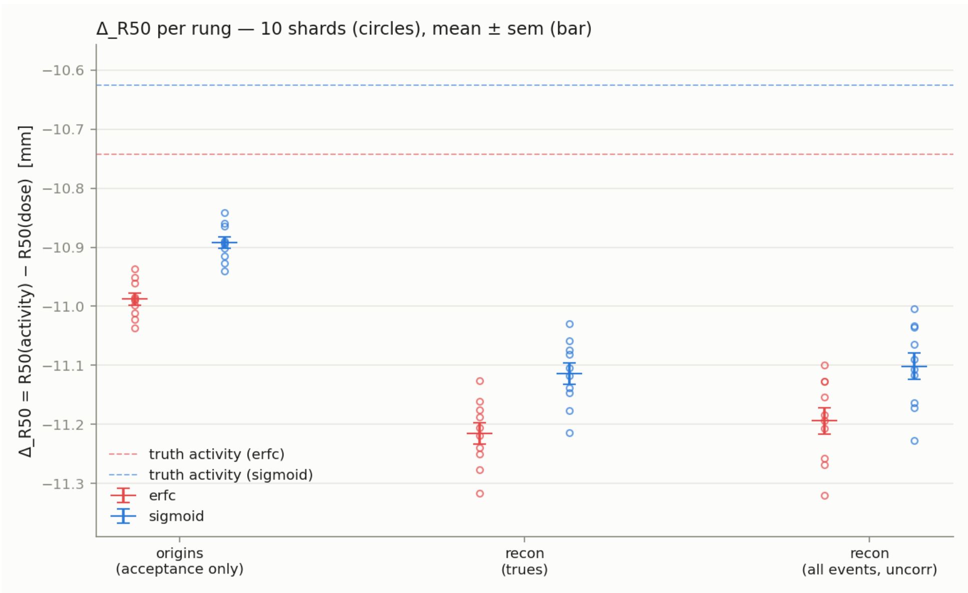

Figure 4: ΔR per rung for the ten acquisitions (circles) with mean ± sem (bars), for the erfc (red) and logistic (blue) fits; the dashed lines are the truth-level values. The two models differ by a constant offset and agree on every shift to 0.02 mm.

The acceptance stage moves the edge by −0.25 mm, the reconstruction by a further −0.23 mm, and admitting the scattered and random coincidences uncorrected changes it by 0.02 mm. The total, −0.45 mm, is the constant the reference simulation calibrates; each term is measured to 0.01 mm to 0.02 mm, far below the effects the verification aims to detect.

The precision is quoted as the spread over independent acquisitions — the fluctuation a real measurement has. (The per-fit errors on reconstructed profiles are conservative by a factor ≈ 2.5, a known consequence of the correlations MLEM introduces between neighbouring voxels.) A single 1 Gy acquisition on this scanner measures the distal edge with

$$\sigma_R(1\,\mathrm{Gy}) \;=\; 0.057(13)\,\mathrm{mm}. \tag{4}$$

4 Scattered coincidences

The scattered coincidences (18.7% of the events) are the one background with structure, and their effect is a scanner property, so it belongs in the comparison budget. Reconstructing the scatters of one acquisition alone (Fig. 5) shows what they contribute to the joint image: a smooth dome peaked upstream of the edge, at 7% of the plateau inside the fit window, falling at $-2.2(4)$ per mm across it. If the fit absorbed none of that slope, R_{50} would shift by -0.20 mm; the measured shift (Table 1) is 0.02 mm, because the baseline and width of Eq. (2) absorb the smooth component. The scatter term therefore needs no correction at the present precision.

The margin is configuration-dependent, and this is where the crystal matters: the CsI variant of the scanner [4] improves the energy resolution well beyond the 10% FWHM of BGO, cutting the scatter fraction directly at the 511 keV window. The scatters-only reconstruction is the standing per-configuration check: the trigger for a correction is a window slope no longer small against the edge gradient.

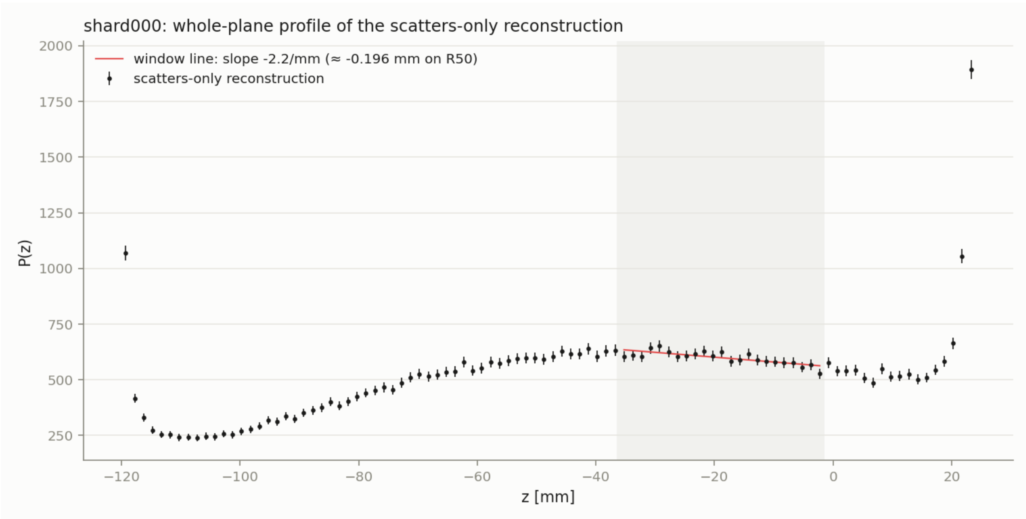

Figure 5: Whole-plane profile of the scatters-only reconstruction of one acquisition, in the units of the joint (all-events) image. The red line is a straight fit inside the window; its slope bounds the worst-case effect on R_{50} at -0.20 mm, against a measured 0.02 mm.

5 Low-dose operation

The dose axis is explored by thinning each stored acquisition with the appropriate Bernoulli probability and reconstructing each realisation with the identical chain. Figure 6 shows ΔR for every realisation, grouped by dose, and the measured σ_R against the $1/\sqrt{\text{dose}}$ law anchored at 1 Gy. Two facts emerge, and Table 2 quantifies the two operating points:

Table 2: The two operating points (erfc fit, whole-plane profile, trues). n is the number of independent realisations.

dose [Gy]	n	ΔR [mm]	sem [mm]	σ_R [mm]	$\pm$ [mm]
1.0	10	−11.216	0.018	0.057	0.013
0.1	20	−11.248	0.040	0.177	0.029

First, the calibration is dose-invariant: the group means agree with the 1 Gy anchor within one standard error down to 0.1 Gy — the reconstruction places the edge at the same depth over

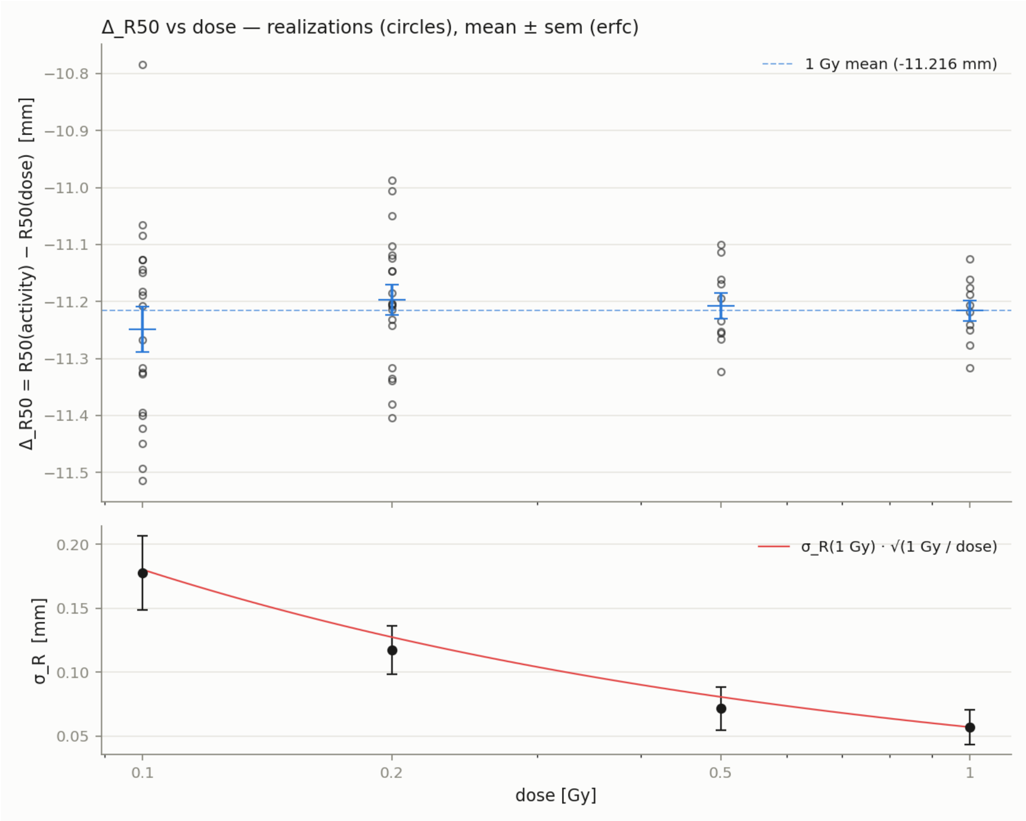

Figure 6: Top: ΔR for every realisation (circles) grouped by dose, with mean $\pm$ sem; the dashed line is the 1 Gy mean. Bottom: the spread σ_R per dose against the $1/\sqrt{\text{dose}}$ prediction anchored at 1 Gy (red line).

a factor ten in statistics. Second, the precision is purely statistical: $\sigma_R \propto 1/\sqrt{\text{dose}}$ at every measured point, with no onset of fit instability at low counts.

The consequence is a mode of operation: an *exploratory dose*. A 0.1 Gy shot — a small fraction of a 1 Gy to 2 Gy daily fraction — locates the distal edge with $\sigma_R = 0.18$ mm and a calibration known to 0.04 mm, an order of magnitude below the millimetre scale that drives clinical margins [5]. The beam range can therefore be verified against the plan before the therapeutic dose is delivered, with the same scanner and the same analysis; extrapolating the measured scaling, even 0.05 Gy would still reach ≈ 0.25 mm.

6 Outlook

Three directions follow directly from this study.

Acquisition start time. The isotope activities (Fig. 7) set the trade-off of in-room imaging: a start 2 min to 3 min after beam-off, rather than one, loses part of the ^{15}O signal but shifts the mix towards ^{11}C, whose positron endpoint energy (0.96 MeV, against 1.73 MeV for ^{15}O) carries a correspondingly smaller positron range — a *sharper* intrinsic edge. A high-sensitivity scanner can absorb the count loss, so the delayed start may improve the range measurement rather than degrade it. This will be quantified as a pure event-time selection on the stored acquisitions.

Isotope-resolved edge model. The distal fall-off is a superposition of per-isotope edges with known relative depths. A composite erfc with simulation-fixed offsets and free amplitudes turns the edge shape into a measurement of the isotope mix, and is the natural refinement of Eq. (2) for the start-time studies.

Scanner comparison. The analysis of this note — the calibration budget, the scatter check and the dose scan — runs unchanged on any configuration of the products tree. The CsI arm and the open geometries are the next entries in the comparison, on the axes where they differ: sensitivity, energy resolution (the scatter term of Sec. 4) and angular coverage.

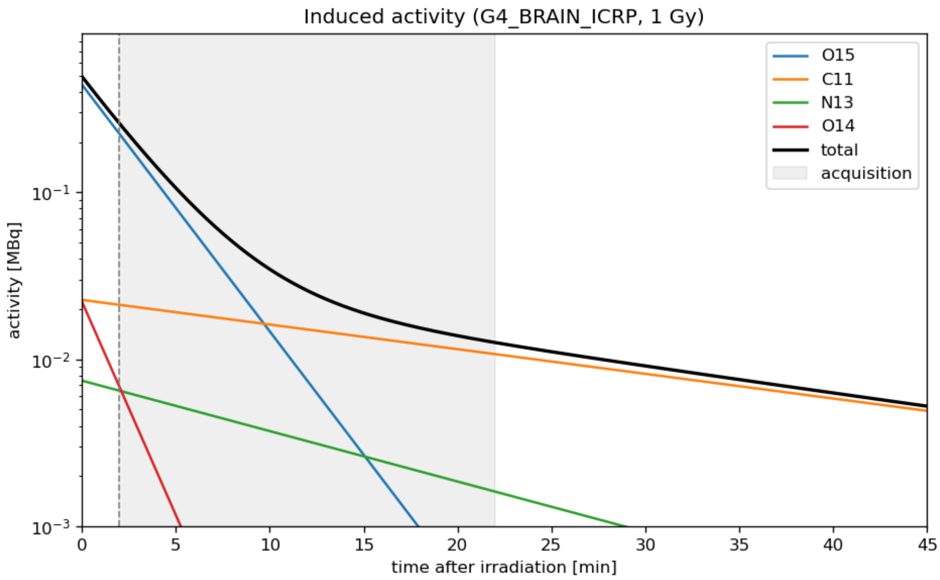

Figure 7: Activity of the β^+ emitters produced by the field, as a function of time: the currency of the acquisition-start trade-off.

Conclusions

The whole-plane erfc fit of the β^+ depth profile, with R_{50} as the range observable and the activity–dose offset calibrated on the reference simulation, is a robust distal-edge estimator: its conventions contribute below 0.1 mm, its calibration budget totals 0.45 mm with every term measured to 0.01 mm to 0.02 mm, and it needs no scatter correction at the present precision. On the reference closed-ring CRYSP scanner it measures the distal edge with $\sigma_R = 0.057$ mm at 1 Gy and 0.18 mm at 0.1 Gy, with a dose-invariant calibration — precision at the tenths-of-a-millimetre level, and an exploratory low-dose verification shot as a direct clinical corollary.

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